

$$\Delta N = -\lambda N \Delta t$$

where λ is a constant of proportionality (decay constant).

Without any further assumptions, this leads to the exponential radioactive decay result:

$$N = N_0 e^{-\lambda t}$$

Radioactive Decay Constant:

The rate of radioactive decay is typically expressed in terms of either the radioactive half-life, or the radioactive decay constant. They are related as follows:

$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda} \approx 0.693\tau$$

The diagram shows the equation $T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda} \approx 0.693\tau$. Below the equation, there are three boxes with labels: 'Radioactive half-life' (pointing to $T_{1/2}$), 'Radioactive decay constant' (pointing to λ), and 'Mean lifetime' (pointing to τ).

The decay constant is also sometimes called the disintegration constant. The half-life and the decay constant give the same information, so either may be used to characterize decay. Another useful concept in radioactive decay is the average lifetime. The average lifetime is the reciprocal of the decay constant as defined here.

For example, free neutrons decay with a halflife of about 10.3 minutes. This corresponds to a decay constant of .067/min and an average lifetime of 14.8 minutes or 890 seconds.

For geologic dating processes involving long half lives, like potassium-argon dating, the decay constant is typically expressed in yr^{-1} and the half-life in years.